



VentCompany

standard_products.py	(, , , ,)
metal_cutting.py	
auto_specification.py	, , JSON/CSV/TXT/HTML
db_integration.py	SQLite: , , ,
demo_usage.py	

1.

.py ventilation_company/ :

```
ventilation_company/
├─ __init__.py
├─ config.py
├─ database.py            ← ваша існуюча БД
├─ gui.py
├─ cad_editor.py
├─ camduct_editor.py     ← можна замінити/доповнити metal_cutting.py
├─ detail_calculator.py
├─ standard_products.py   ← НОВИЙ
├─ metal_cutting.py       ← НОВИЙ
├─ auto_specification.py   ← НОВИЙ
├─ db_integration.py       ← НОВИЙ
├─ project_builder/
|   ├── components.py     ← можна доповнити з standard_products.py
|   ├── specifications.py ← можна замінити/доповнити auto_specification.py
|   └─ ...
```

2. (standard_products.py)

:

- RectFlange —
- RoundFlange —
- RectTee —
- RoundTee —
- RectTransition —
- RoundTransition —
- RectElbow —
- RoundElbow —
- RectCap —
- RoundCap —

- `make_rect_duct()` —
- `make_round_duct()` —

:

```
from ventilation_company.standard_products import (
    ProductLibrary, RectFlange, MaterialType, Thickness
)

lib = ProductLibrary()

lib.add(RectFlange(
    name="Фланець 400×200",
    width=400, height=200, length=0,
    thickness=Thickness.T1_0,
    material=MaterialType.GALVANIZED,
    quantity=10
))

print(f"Площа: {lib.get_total_metal_area():.3f} м²")
print(f"Вага: {lib.get_total_weight():.3f} кг")
```

3. (metal_cutting.py)

:

- Bottom-Left heuristic —
-
- 90°
-

:

```
from ventilation_company.metal_cutting import MetalCutter

products = [...] # список виробів зі standard_products
cutter = MetalCutter(sheet_width=1250, sheet_height=2500, thickness=0.7)
plan = cutter.calculate_from_products(products)

print(f"Листів: {plan.total_sheets}")
print(f"Використання: {plan.overall_utilization*100:.1f}%")
```

4. (auto_specification.py)

:

-
- , ,

-
-
- : JSON, CSV, TXT, HTML

:

```
from ventilation_company.auto_specification import SpecBuilder

builder = SpecBuilder(project_name="Мій проєкт")
builder.set_material_price("оцинкована сталь", 55.0)

for product in products:
    builder.add_product(product)

spec = builder.build()
builder.save_to_file("spec.html", format="html")
```

5. (db_integration.py)

, :

- projects —
- project_products —
- specifications —
- cutting_plans —
- standard_products_library —
- material_prices —
- clients —

:

```
from ventilation_company.db_integration import ProjectDatabase

db = ProjectDatabase("data/company.db")

# Створити проєкт
project_id = db.create_project(name="Вентиляція А", client="ТОВ "Будівельник")

# Додати виріб
db.add_product_to_project(project_id, {
    "name": "Фланець 400×200",
    "product_type": "фланець прямокутний",
    "width": 400, "height": 200, "length": 0,
    "thickness": 1.0, "material": "оцинкована сталь",
    "quantity": 10,
    "metal_area_m2": 0.234,
    "weight_kg": 1.84
})

# Зберегти специфікацію
```

```
db.save_specification(project_id, spec_data=spec.to_dict())

# Отримати звіт
report = db.get_production_report()
```

6. ()

```
from ventilation_company.db_integration import save_project_full

result = save_project_full(
    project_name="Вентиляція офісу",
    products=products_list,
    spec_data=spec.to_dict(),
    cutting_plan=plan.to_dict()
)

print(f"Проект ID: {result['project_id']}")
```

7. GUI

gui.py / :

```
from ventilation_company.standard_products import ProductLibrary
from ventilation_company.auto_specification import SpecBuilder
from ventilation_company.db_integration import ProjectDatabase

class MyGUI:
    def __init__(self):
        self.lib = ProductLibrary()
        self.db = ProjectDatabase()

    def add_flange(self, w, h, qty):
        from ventilation_company.standard_products import RectFlange, Thickness, MaterialType
        self.lib.add(RectFlange(
            name=f"Фланець {w}×{h}",
            width=w, height=h, length=0,
            thickness=Thickness.T1_0,
            material=MaterialType.GALVANIZED,
            quantity=qty
        ))

    def generate_spec(self):
        builder = SpecBuilder(self.current_project_name)
        for p in self.lib.to_dict():
            builder.add_product(p)
        return builder.build()
```

```
def save_to_db(self):
    spec = self.generate_spec()
    self.db.save_specification(self.project_id, spec.to_dict())
```

8.

```
cd ventilation_company
python demo_usage.py
```

9.

Python:

- sqlite3
- json
- csv
- math
- dataclasses
- typing
- collections
- datetime
- html
- io
- copy

pip- !